

Pharmacies – Journey of online purchase

A Service Design Project guided by Universal Design

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1. Introduction & project overview

This project focuses on the digital service journey of purchasing medicines through online pharmacies. Currently, complex interfaces, mandatory e-identification like BankID, and information overload create significant accessibility barriers, leaving many users feeling digitally excluded. Guided by Service Design and Universal Design principles, the overall aim of this project is to rethink and redesign this process to ensure safe, equitable, and inclusive access for all users.

2. Context, theory, and in-depth perspectives

2.1 The service, context and societal relevance

The service focuses on the online purchasing of medicines through pharmacy platforms, encompassing product browsing, identification (e.g., BankID), prescription verification, and payment. While online pharmacies hold societal promise by mitigating geographical barriers and enabling independent access to healthcare, their complex digital interfaces currently exacerbate exclusion. National surveys reveal that only about 50% of people with disabilities feel digitally included (PTS, 2021; SMFOI, 2021). This exclusion creates an economic burden, as marginalized groups are often forced to pay higher in-store prices (Svensson, 2024).

This project intersects with the European Accessibility Act, which mandates accessible e-commerce, and EU eIDAS regulations (European Commission, 2025). Accessible pharmacy designs directly support UN Sustainable Development Goals 3 (good health) and 10 (reduced inequalities) by promoting equity and autonomy (United Nations, 2015). Conversely, Sweden's near-monopoly of BankID sidelines non-users, conflicting with social sustainability aims (Copenhagen Economics, 2024).

Our research identified critical barriers: opaque multi-step processes, disjointed payment integrations, minuscule text, and fast-expiring QR codes. These issues are acute for screen-reader users or those lacking e-ID access. Background research across countries

like Japan, Finland, and Canada highlights that while identification hurdles are global, adopting inclusive authentication like biometrics and guest checkouts can significantly reduce reliance on a single technology (BPOM, 2022; Fimea, n.d.)

2.2 Using a service design approach guided by Universal design principles

Service design employs human-centered, iterative methods to holistically map and reshape services across journeys, stakeholders, and front-/backstage elements, transcending isolated interfaces (Stickdorn et al., 2018). Stickdorn et al. (2018) underscore its collaborative visualization for uncovering systemic frictions, crucial here as pharmacy issues surface in transitions (e.g., pharmacy site to e-ID/payment) (see fig 2.1).



Fig. 2.1 Online Pharmacy User Journey Map

Universal Design (UD) proactively crafts products, environments, and services usable by the widest array of people possible, sans adaptation or specialization (Connell et al., 1997). Evolving to its "third generation," UD rejects norm-deviation binaries, where designs cater to an imagined "normal" user with add-ons for "deviants" favoring "nonclusive design": solutions eschewing body/role categorizations, embracing contextual variation, and enabling fluid appropriation without predefined user limits (Hedvall et al., 2022). Nonclusive UD shifts from explicit accommodations (e.g., wheelchair icons) to tacit, convergent features (e.g., adjustable counters signaling function over identity), fostering proactive, aware processes that normalize diversity via intersectional lenses (Hedvall et al., 2022). For this project, it reframes pharmacy flows as variation-supportive: screen-reader/keyboard-compatible,

multi-option ID/payment, low-complexity challenging "security = smartphone BankID" to prioritize privacy/autonomy universally (SMFOI, 2021).

2.3 In-depth perspectives from assignment 1

Based on Assignment 1, we translated our distinct disability advocacy roles into research-backed user need statements (see fig 2.2).

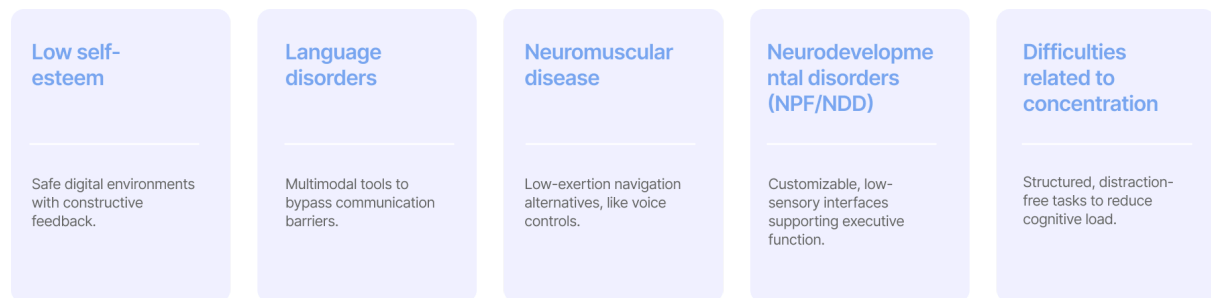


Fig 2.2 User needs statements formulated based on our advocacy areas

By anchoring these diverse cognitive, emotional, and physical needs in empirical research, we ensure our personas avoid reductive stereotypes and accurately reflect real user requirements.

2.4 First reflection on expectations, social considerations, and the arguments for working towards accessible design

Our primary expectation was achieving a "curb-cut effect"—designs benefiting all users. We were frustrated by inaccessible online services and eager to address overlooked cognitive barriers versus physical ones.

Current designs stigmatize by embedding "ideal user" norms (tech-savvy, BankID-proficient), marginalizing others as deficient. Non-users face costlier stores or dependency, evoking discrimination (SMFOI, 2021; Svensson, 2024). Hedvall et al.'s (2022) nonclusive design counters this via equal multi-paths (web flows, alt e-IDs, assisted channels), normalizing variation and reducing shame.

Accessibility creates curb-cut benefits: clearer pharmacy steps/language aid parents, novices, and low-bandwidth users broadly (Hamraie, 2017). It enhances UX, trust, efficiency, equity, and demographic future-proofing while meeting ethical/business goals (Copenhagen Economics, 2024; United Nations, 2015).

3. Automated accessibility testing

3.1 Automated digital accessibility testing

We evaluated Apoteket, a widely used online pharmacy, using SilkTide and Attention Insight to gain complementary accessibility insights. SilkTide assessed technical accessibility aspects such as color contrast, screen reader compatibility, focus order, alt text, headings,

and simulations for visual impairments and dyslexia. Attention Insight analyzed how the visual layout influences user attention through heatmaps. The evaluation focused on key pages in the purchasing journey: the homepage, search results, product details, and cart.

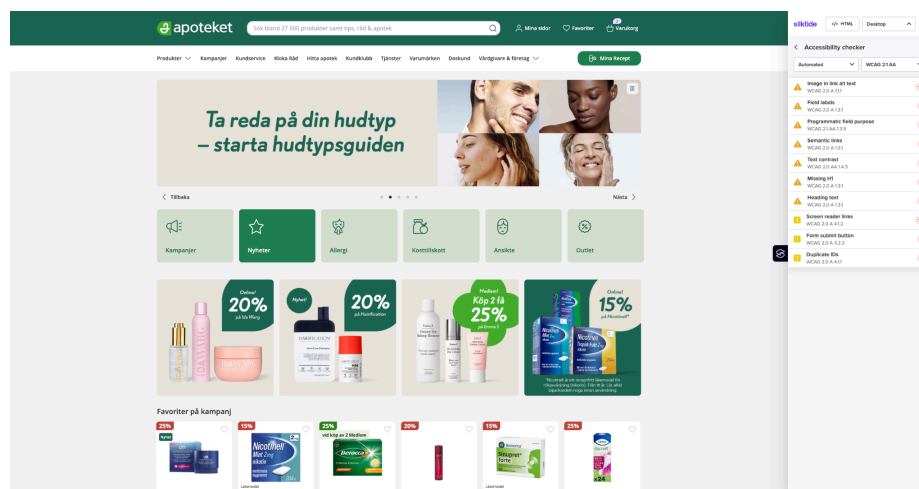


Fig. 3.1 Automated accessibility testing with SilkTide

SilkTide identified several technical accessibility issues, including missing alt text, unclear or empty links, missing headings, undefined form field purposes, and insufficient color contrast (see fig 3.1). These issues directly affect how assistive technologies, such as screen readers, interpret and present content.



Fig. 3.2 Automated accessibility testing with Attention Insight

Attention Insight complemented this analysis by examining visual attention through heatmaps (see fig 3.2). The results showed that banners and promotional content attract the most attention. While this may be beneficial from a business perspective, it can distract users from essential elements such as the search bar.

3.2 Reflection

Based on our perspective within our advocacy area, several issues were identified as follows (see fig 3.3)

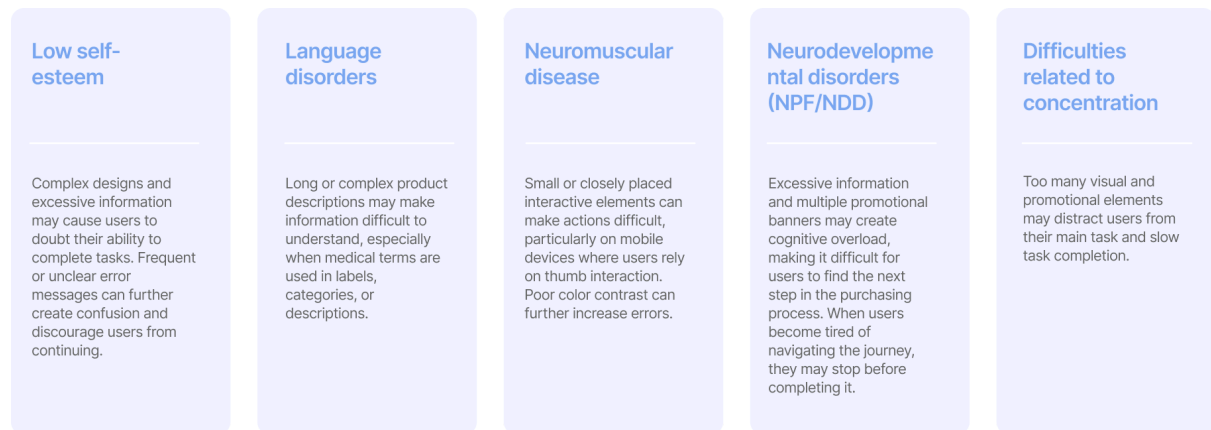


Fig 3.3 Key accessibility issues within our chosen advocacy areas.

Looking at the results, not all accessibility issues can be identified by automated accessibility tools. However, they cannot directly identify behavioral or psychological issues. While some findings may support our interpretation of potential accessibility problems, automated tools alone cannot fully explain how users experience the interface. Human feedback and user-centered evaluation are still required to better understand how the design affects users when completing their tasks.

4. Insights from lived experience

4.1 Consultation preparations

Before the consultation, our group prepared and printed a set of questions focusing on participants' experiences using online pharmacies. We also prepared an information sheet and consent form in both English and Swedish to ensure that participants could clearly understand the purpose of the consultation and their rights. When designing the questions, we avoided overly complex language to make them accessible and easy to understand. Roles were divided within the group prior to the consultation. Since the interviews were recorded, we chose not to take detailed notes during the session so that we could focus fully on the discussion. The Swedish-speaking students facilitated the interviews conducted in Swedish, while the remaining interviews were facilitated by the other group members.

4.2 Initial insights and next step

After the consultation, our group discussed the interviews and identified several initial insights. One recurring theme was that many participants prefer visiting physical pharmacies because they can receive direct help and guidance from staff. Another observation was that online pharmacy websites can be cognitively demanding. Some participants mentioned that there is a large amount of information, such as campaigns or moving banners, which can create information overload and make it difficult to focus on the intended task. Participants also described challenges with navigation, sometimes clicking through several buttons before finding the correct option.

Before further analysis of the data, each group member was tasked with transcribing a recording from the consultations. This was done using [AI] and later refined and corrected by the assigned group members. The Swedish interview transcripts were also translated by Swedish-speaking group members to make sure they were accessible to the rest of the group.

4.3 Critical reflection on roles, power, and ethics

Designers should consider accessibility of communication, ethics and power dynamics. They have the responsibility to create an inclusive and respectful environment where participants feel safe and comfortable sharing their perspectives and experiences. Communication should be clear and accessible, which includes using simple language, avoiding speaking too quickly and ensuring that consultants understand the questions being asked. Facilitators should also allow extra time for responses when needed. The primary goal of the designers is to listen and learn from the participants' experiences instead of dominating the conversation. Designers should also avoid making them feel like research subjects and ensure that participation is voluntary and based on mutual respect.

According to Arnstein's Ladder of Participation (see fig 4.1), the involvement of the consultants in our project can be situated at the level of consultation. The consultants were invited to share their experiences during the initial interviews and later provide feedback on our low-fidelity prototype in an online meeting. While their perspectives helped inform and improve our design ideas, the final design decisions were made by the group instead of the consultants themselves.

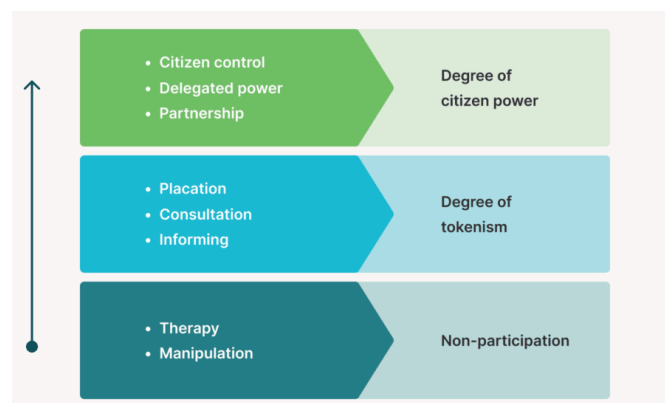


Fig. 4.1 Arnstein's Ladder of Participation

Terms used for people involved in accessibility work shape how they are perceived in the design process. Calling them "expert users" values their lived experience and knowledge, while "extreme users" can frame them as outside the norm. Referring to participants as "consultants" emphasizes respect and collaboration, recognizing their insights as important contributions to design decisions.

5. Defining problems and formulating accessibility requirements

5.1 Analysis and clustering of insights

To formulate requirements and define the problem area of the project, the consultation interviews needed to be analysed. This was done by transcribing and coding the recorded interviews using a method of inductive coding, where codes were created based on the data. We performed the analysis using independent coders, taking care that Swedish-speaking group members coded the Swedish interviews, to reduce the risk of misinterpretation due to incorrect translation.

Once codes had been created, they were grouped together to create themes using Affinity Diagramming. A shared Miro board was used with the whole group present. Each person individually clustered together codes, before collaboratively going through each cluster to make sure they represented a shared idea of the problems (see fig 5.1).



Fig. 5.1 Clustering together themes into problem statements



Fig 5.2 Final problem statements arranged in order of importance

The final results of the Affinity diagramming and overall data analysis showed us five main focus areas with which we could continue (see fig 5.2). The different areas were arranged in order of importance based on the number of times it was brought up during consultations, the risk, and the scope of the consequences.

5.2 Formulated accessibility requirements

To continue with ideation, we needed to create some clear, concise accessibility requirements. This was done by creating user need statements which would be able to guide ideation forwards and focus on needs of the users, rather than the problems found. These were formulated based on the focus areas previously found.

1. Disabled users *need* simple, easy-to-navigate online pharmacies to avoid cognitive barriers when ordering medication.

2. Disabled users *need* personalized guidance to understand medications and the ordering process.
3. Disabled users *need* a quick and efficient way to order medicine *because* convenience and time-saving are key reasons they choose online pharmacies.
4. Disabled users *need* a way to easily compare prices between pharmacies *because* price is a key factor when deciding where to purchase their medication.
5. Disabled users *need* secure and reliable delivery options, including pick-up points, to feel confident receiving medication.

6. Ideation and conceptual Service design

6.1 Ideation, prioritisation and design reasoning

We used brain writing and brain dumping methods to generate ideas collaboratively. In brain writing, each member wrote ideas individually before sharing, while brain dumping allowed rapid collective idea capture. This approach ensured diverse perspectives from our advocacy areas without dominance by vocal members.

Five key concept directions emerged from the ideation session (see fig. 6.1). First, step-by-step navigation to slow down and structure the ordering process. Second, a humane guidance model inspired by the receptionist pattern, combining an interactive video bot with the option to connect to a real pharmacist. Third, a standardized cross-pharmacy experience to reduce the need for users to relearn different site structures. Fourth, clearer categorization, separating medicine from other pharmacy products. Fifth, a more visual browsing approach, including body-part-based navigation for users unfamiliar with medical terms. These directions aligned with usability principles such as recognition rather than recall, consistency and standards, match between system and the real world, and aesthetic and minimalist design.

To narrow the ideas down, we used a lightweight decision matrix inspired by service design prioritisation methods. In our case, we compared ideas according to three main criteria: their expected accessibility impact, their alignment with Universal Design principles, and their feasibility in relation to realistic constraints such as time, technology, and implementation complexity. We also reflected on existing examples from other contexts, such as pharmacy and medication services in other countries, to judge whether some of the ideas were conceptually and technically plausible.

This prioritisation process helped us distinguish between supporting interface improvements and more central concept directions. Changes such as fewer banners, larger text, and clearer hierarchy remained important design principles, but they were not enough on their own to address the systemic barriers identified in the journey. As a result, we chose to move forward not with one isolated feature, but with a combination of ideas that together could reduce cognitive load, improve confidence, and create a more coherent service experience. To develop these directions further, we then used parallel low-fidelity prototyping in Figma and Balsamiq, with two groups exploring alternative versions of the concept before combining the strongest elements into one shared direction. Parallel low-fidelity prototyping

also fits this stage well, since such prototypes are useful for comparing alternative solutions and testing broad concepts before committing to one final structure.

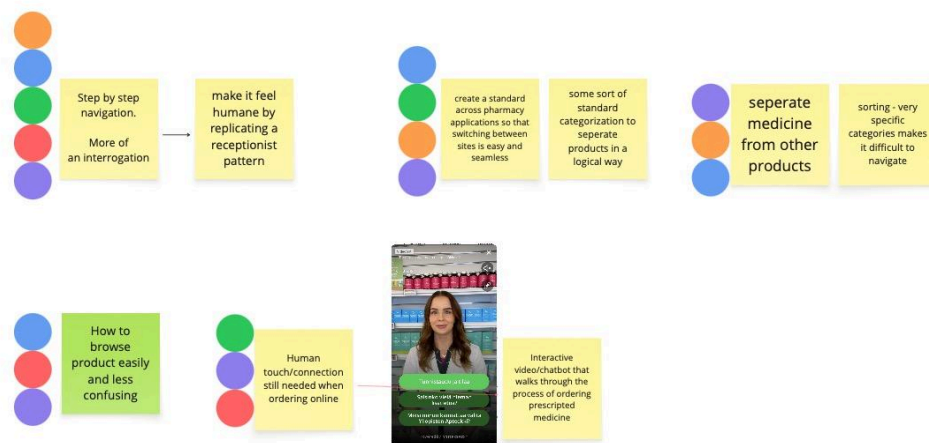


Fig 6.1 Miro ideation board illustrating top ideas prioritized for low-fidelity prototyping.

6.2 Design concept version 1

We selected a unified pharmacy app that acts as a centralized aggregator and shopping assistant, bringing together multiple pharmacies, support modes, and navigation paths within one service. Rather than solving a single page or feature, the concept addresses the journey as a whole: where to start, how to navigate, how to compare options, and how to complete the purchase with reassurance similar to a physical pharmacy.

The app combines our strongest directions: frictionless checkout with an integrated wallet and saved addresses, built-in price and delivery comparison, and three flexible interaction profiles: Live Help, AI Help, and Self Help. Users can switch profiles at any point, since no single interaction model fits all users or all situations. Even in Self Help, multiple navigation modes (e.g., visual, voice, simplified flows, body-part-based browsing) reduce cognitive load and reliance on medical terminology.

This concept primarily supports users who experience cognitive overload, concentration difficulties, language-related barriers, low self-esteem, or insecurity when ordering medicine online, while also benefiting older adults, caregivers, and first-time or stressed users through a curb-cut effect. Clearer structure, optional support, and more intuitive categorization improve usability without singling out any one group. Guided by Universal Design and nonclusive design, we focused on equitable use, simple and intuitive operation, perceptible information, and flexible pathways while remaining aware that, if implemented poorly, too many options or an overbearing AI could introduce new forms of confusion or mistrust.

7. Evaluation and Iteration

7.1 Evaluation approach

The evaluation was conducted online through Zoom calls. During the session, we explained our concept using a prototype that we had developed. The prototype only included the main functional parts so that the idea could be presented in a simple and easy-to-understand way. We controlled the interaction with the prototype during the walkthrough to keep the discussion focused and to avoid confusion for the consultants when navigating the prototype themselves. Through this process, the consultants were able to provide feedback while we explained the features and design.

The purpose of the session was to understand whether the proposed features were relevant and potentially helpful for them. We also asked for feedback on whether the visual design felt accessible and clear. The feedback and perspectives in this evaluation were provided by two consultants, Tess and Frida.

7.2 Key insights from the evaluation

The consultants found our idea interesting and helpful. They mentioned that the features address several difficulties related to cognitive load, such as dealing with too much information, navigating many different websites, and going through complex checkout processes. The medical shopping assistant we designed helps users purchase medical products through support from either a human assistant or an interactive bot. This support allows users to communicate easily without needing to understand complex medical terms when searching for medicine.

They also appreciated the product exploration feature using a human body figure, which helps users find products even when they do not know the exact name of the body part or medical condition. During checkout, users do not need to visit multiple websites because the system recommends the best product options based on price and delivery time across different platforms. One consultant mentioned that this idea could help users with insecurities feel more comfortable when buying medicine, since the bot guides them step by step without making them feel judged if they are unsure or need help. Additionally, the consultants found the multiple profile feature useful, as it allows users to purchase medicine on behalf of others, such as their parents.

Although the concept was considered helpful, the consultants suggested several improvements related to the visual design. For example, a language selection feature is needed because not all users understand the same language. Some text elements were also considered less readable and should be slightly larger. In addition, a few icons were not sufficiently representative, which caused confusion about the function of certain buttons. Based on this feedback, we made several adjustments to the interface design. We added a language selection feature, increased the text size to improve readability, and replaced some icons with more representative ones.

7.3 Reflection on your facilitation of the evaluation

Facilitating the evaluation helped us understand how important it is to clearly guide participants through a prototype so they understand the concept and its intended features. By controlling the interaction with the prototype during the walkthrough, we were able to focus the discussion on the core features and receive more targeted feedback. We also

learned how important interface clarity is, especially when designing for users who may already experience cognitive overload.

The evaluation mostly went as planned. However, because the prototype was not fully interactive, we had to explain some functions instead of demonstrating them. In future evaluations, we would use a more interactive prototype and prepare more specific follow-up questions to gather deeper insights.

8. Final concept

The final service concept, based on accessibility requirements and consultations, is an aggregator app which centralizes all your medication needs in one place. This stems from expressed difficulties in understanding how each of the different pharmacies work, and a need for seamless integration with payment options and price comparisons.

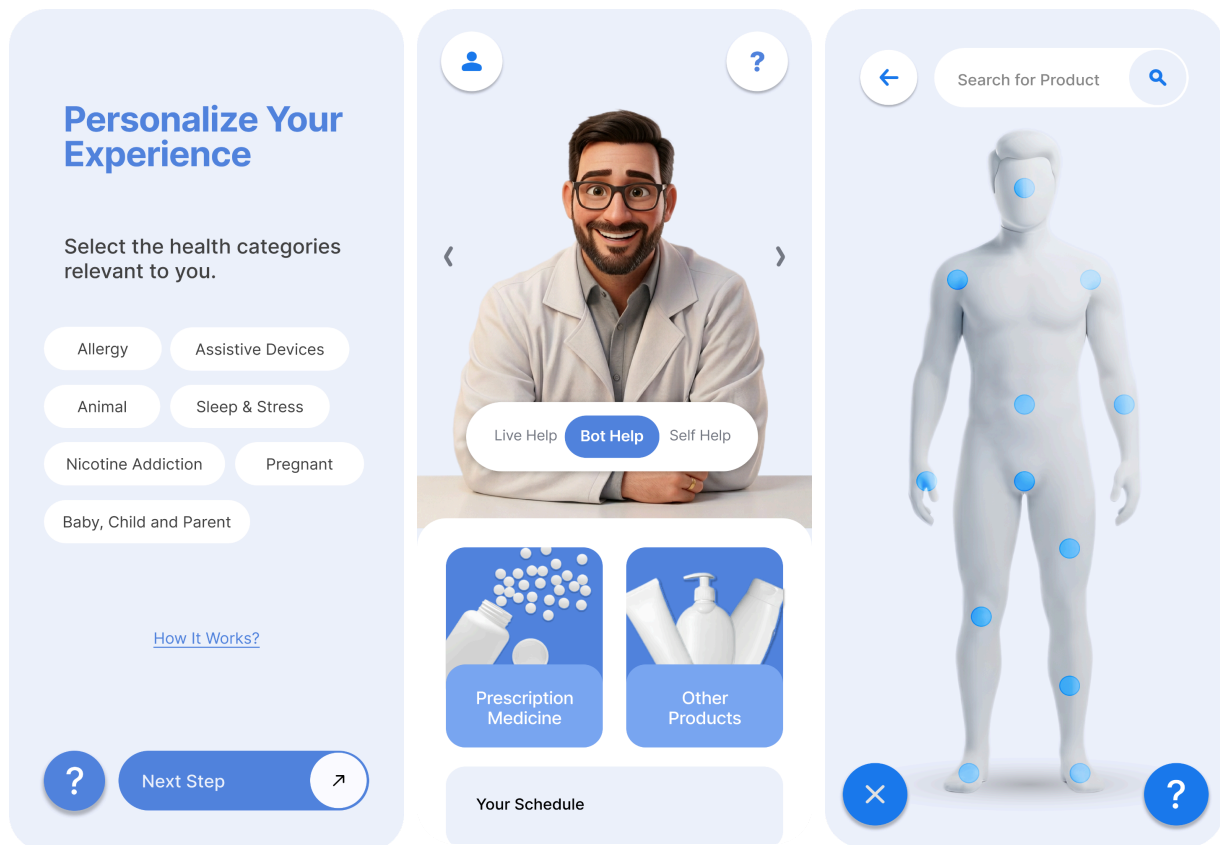


Fig 8.1 Onboarding. Choosing which categories are relevant to you

Fig 8.2 Receiving help from AI bot

Fig 8.3 Human model enables choosing category based on body part

The app starts with a step-by-step onboarding process in which the user is able to personalize their pharmacy shopping experience to suit their needs and wants. This includes choosing categories which are relevant to you (see Fig 8.1), setting up a digital wallet and choosing preferred integrated payment options, and the option to set up multiple profiles (especially for those with children, or those acting as caregivers).

Inside the app, some important separations have been made. Firstly, insights from consultations showed that separating medication from other products found at the pharmacy was important to create an easy-to-navigate interface and reduce confusion and frustration with users. Further, there were mixed preferences in the amount of help/communication users wanted/needed during their online journey. The app therefore includes three different “help modes”. Firstly, you can choose to receive no help and navigate the app by yourself (as is the standard on current pharmacy websites and apps). Secondly, you can choose to receive help from an AI bot which operates through a digital humanoid form (see Fig 8.2). This allows the users to freely ask questions and communicate their needs verbally. Finally, the app also allows the user to receive real-time human help through video calls and chats with actual pharmacists. This is intended to replicate the personable experience found in physical pharmacy stores.

One issue frequently brought up during consultations was the inherently confusing and complicated categorization and navigation when browsing and searching for products. To counteract this, as mentioned previously, we created an intuitive, and body-part focused categorization which started by separating medication from other products (see fig 8.2). While looking at other products, categorization was focused on purpose and body part, for example, separating self care and grooming from sickness relief, and later separating it further into categories like haircare, skincare, etc.

Another feature we decided to add was the human model which allows you to visually choose a part of the body for which you are looking for products (see Fig 8.3). While previously mentioned attempts at simplifying categorization proved to be an improvement during consultations, we acknowledge there is still risk for confusion and misinterpretation. We therefore decided to add a visualization of categorization in order for users to pick their products in a more intuitive way.

Finally, we added a feature which allows users to compare prices and delivery times between different pharmaceutical companies. The idea is that users are able to choose which pharmacy provides the best value for them (price, or speed of delivery) in one place, while also being able to keep the ordering process within the same app.

8.1 Limitations and future considerations

While the final concept is constrained by time and resources, it's important for us to acknowledge its limitations and areas for improvement.

Not every need we identified was equally visible in the concept. For example, delivery and pick-up were important concerns in the earlier analysis, but these were less central in the final design than navigation, guidance, and comparing options. Another limitation is that the concept does not fully solve wider exclusion issues, such as authentication barriers or dependence on existing digital systems.

There are also clear feasibility issues. Firstly, it may be unrealistic to expect a pharmacist to be on call for users in need of help, or. Secondly, we have to acknowledge the privatization of pharmacies in Sweden, meaning that each pharmacy is its own company with their own goals. It would therefore be difficult to implement this aggregator app without the help of legislation and guidelines.

From a Universal Design perspective, one important reflection is that the concept still depends on a digital service environment that may already exclude some users. Although the concept tries to reduce confusion, offer flexibility, and support different needs, it still assumes access to digital devices, digital confidence, and systems that are not inclusive by default. For that reason, this concept is a meaningful step toward a more accessible service journey, but not as a complete Universal Design solution.

9. Conclusion

Throughout this project, several insights emerged from the different phases of the service design process. Automated accessibility testing revealed technical accessibility issues such as missing alt text, unclear link structures, and visual elements that distract from essential functions. Consultations with users highlighted that online pharmacy services are often cognitively demanding due to complex navigation, excessive information, and fragmented processes across multiple platforms. Consultants also emphasized the value of personalized help found in physical pharmacies and expressed concerns about delivery options, pricing, and understanding medication information. These findings informed the development of accessibility requirements focused on simplifying navigation, reducing cognitive load, and providing clearer guidance during the purchasing process.

Based on these insights, our design concept finalized as an aggregator platform that allows users to access products, prices, and delivery options across pharmacies in a single interface. Integrating multiple assistance modes, including independent browsing, AI-supported guidance, and access to human pharmacists, could replicate the supportive experience of physical pharmacies while maintaining the efficiency of online services.

Despite these improvements, several barriers remain. Implementing real-time human assistance may be resource-intensive, and collaboration between competing pharmacy companies could be difficult in practice. Future work should further explore regulatory support, technical feasibility, and continued user involvement to ensure the service evolves toward a more fully realized Universal Design solution.

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